

A Laminate Parallel Articulated Spine for a Cat-Scale Quadruped Robot

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Abstract

This project presents the design and physical prototyping of a bio-inspired mechanism intended to serve as the articulated spine of a small quadruped robot with Jansen mechanisms for its four legs. Taking inspiration from the classical Sarrus linkage, we develop a modified planar parallel linkage that connects two rigid 3D-printed chassis plates and is intended to provide vertical support while allowing controlled in-plane bending. The spine is realized in a Smart Composite Microstructures (SCM) format using a five-layer laminate (rigid–adhesive–flexure–adhesive–rigid). Stiff outer layers form the links, while the flexure layer creates passive spring-like joints that generate a restoring torque as the spine bends. The overall morphology is scaled to roughly match the proportions of a juvenile cat while operating at a much lower mass of about 250 g. We construct both a paper analogue of the mechanism and a laminate prototype integrated into a quadruped chassis. A planar kinematic model, implemented in Python, approximates the six-link Sarrus-based spine as an equivalent three-joint serial chain in the sagittal plane. Using this model, we compute representative joint torques and joint velocities for expected ground reaction forces and vertical motions of the front chassis. The results demonstrate that a linkage-based, passively compliant spine is feasible within the constraints of laminate fabrication and provides a structured way to tune the directional stiffness of the robot torso for future integration with Jansen-leg quadruped locomotion.

1 Introduction

Bio-inspired legged robots often borrow structural ideas from animals to improve stability, efficiency, and maneuverability. Many existing quadruped designs use a rigid torso with compliant legs, or a uniformly flexible backbone that bends easily in all directions. While these approaches simplify design and control, they limit the role of the spine to either a passive connector or a uniformly compliant beam, which can reduce force transmission and restrict useful body motion during locomotion.

In contrast, animals such as cats and cheetahs rely on spines that are stiff in some directions and compliant in others, allowing the torso to store and release energy, extend stride length, and stabilize the body during fast gaits. Motivated by this idea, this project explores a mechanical articulated spine for a small quadruped robot driven by Jansen mechanisms at each leg. The design is inspired by the classical Sarrus linkage but modified into a planar parallel structure that connects two rigid 3D-printed chassis plates. The links are made from stiff laminate layers, while the joints are implemented as thin flexures that behave like passive spring-like hinges due to the material properties. The goal is to achieve a spine that provides vertical support, allows controlled in-plane bending, and naturally generates a restoring torque as it deflects.

This report focuses on the mechanism-level design, scaling, and physical prototyping of this spine using laminate construction. Dynamic behavior and full leg–spine coupling are left for future work.

1.1 Project Rationale and Scope

Scope. This project makes the spine the central design element of a small quadruped platform:

1. **Parallel Articulated Spine.** Design and prototype a Sarrus-inspired planar parallel linkage that connects a front and rear chassis, targeting high vertical stiffness with controlled in-plane bending between the two bodies.
2. **Five-Layer Laminate Implementation.** Fabricate the spine links and joints using a five-layer SCM stack (rigid–adhesive–flexure–adhesive–rigid) so that links remain stiff while joints behave as compliant, spring-like flexures.
3. **Geometric Compatibility with Jansen Legs.** Dimension the spine assuming that each chassis will carry a planar, single-degree-of-freedom Jansen leg mechanism that provides a more biologically relevant foot trajectory than simple rotary legs, even though full leg–spine integration is deferred to later work.

The work in this report is limited to geometric design, basic stiffness reasoning, and fabrication of spine prototypes using laminate construction with 3D-printed chassis interfaces.

Impact. By replacing a rigid or uniformly flexible torso with a parallel articulated linkage whose joints behave as passive spring-like elements, this project isolates the spine as a key morphological variable. The prototype provides initial insight into how a Sarrus-inspired, passively compliant spine can transmit loads between the front and rear chassis while still permitting useful bending and self-centering behavior. These results inform future quadruped designs that must balance support, compliance, and integration with complex leg mechanisms such as Jansen linkages.

2 Bio-Inspiration and System Scaling

2.1 Biological Basis and Hybrid Concept

The proposed robot combines two distinct bio-inspired ideas into a single hybrid morphology. At the *body-plan level*, the platform is scaled to a small quadruped, roughly corresponding to a juvenile domestic cat with four primary limbs and a compact trunk. Quadrupeds of this size use their spine to support body weight while allowing controlled bending in the sagittal plane to extend stride length and smooth center-of-mass motion during locomotion.

At the *mechanism level*, the internal backbone draws inspiration from myriapod-style robots that use Sarrus-like linkages to realize segmented, popup-compatible trunks. Prior centimeter-scale “micropod” designs employ Sarrus mechanisms and laminate flexures to create compliant, vertically supportive backbones using only revolute joints. In this project we adapt that idea to a quadruped context: two rigid 3D-printed chassis plates (front and rear) are connected by a Sarrus-inspired parallel articulated spine, implemented in a five-layer laminate stack. Stiff outer layers form the links, while the thin flexure layer creates joints that act as passive torsional springs and generate a restoring torque as the spine bends.

This results in a hybrid system: externally, the robot resembles a cat-sized quadruped with four Jansen mechanisms providing leg motion; internally, the connection between the front and rear bodies behaves like a myriapod-inspired, Sarrus-based articulated spine.

2.2 Scale Selection

The overall geometric and dynamic scale of the robot is chosen to be proportionally similar to a juvenile cat while keeping the total mass much lower to match the capabilities of laminate fabrication and small actuators. Table 1 summarizes the key parameters used for this scaling. The robot body length (distance between front and rear chassis) is set to 0.25 m, with an approximate total mass of 0.25 kg.

Table 1: Key morphological and locomotion parameters for a juvenile cat and the proposed quadruped robot. Biological values are approximate and used only for scaling.

Quantity	Symbol	Juvenile cat	Robot target
Body mass	m	1.5 kg to 2.0 kg	0.25 kg
Body length (shoulder–hip)	L_{body}	0.25 m to 0.30 m	0.25 m
Typical walking speed	v	0.4 m/s to 0.6 m/s	0.2 m/s to 0.3 m/s
Stride length	L_{stride}	0.30 m to 0.40 m	0.20 m to 0.25 m
Step frequency	f_{step}	1.5 Hz to 2.0 Hz	1.0 to 1.5
Max foot height (swing)	h_{foot}	0.03 m to 0.05 m	0.02 m to 0.03 m
Peak vertical GRF per leg	$F_{z,\text{max}}$	20 N to 30 N	3 N to 5 N (design)
Spine flexion range (sagittal)	θ_{spine}	$\approx 10^\circ$	$\approx 10^\circ$

The robot mass is intentionally much lower than that of the biological counterpart, which reduces expected ground reaction forces and laminate stresses. With a mass of 0.25 kg, the body weight is approximately 2.45 N. Designing for peak vertical ground reaction forces of 3 N to 5 N per leg provides a safety margin for dynamic effects while remaining compatible with small actuators and laminate flexures.

2.3 Supporting Literature

Several prior works inform the specific design and fabrication choices in this project. Hoover and Fearing introduced the Smart Composite Microstructures process for laminate robots and demonstrated complex linkage mechanisms fabricated from FR4 and Kapton layers [1]. Eckert *et al.* experimentally explored the coupling between leg design and spine stiffness in small quadrupeds [2]. Bing *et al.* showed that lateral spinal flexion can improve turning behavior and performance in small legged robots [3]. Dürre *et al.* provided a comprehensive survey of legged locomotion and inter-limb coordination for multi-legged animals, which helps motivate our long-term goal of integrating the articulated spine into a biologically grounded gait framework [4]. The textbook *Foldable Robotics* by Aukes [5] provides the lamination and hinge design rules used in our five-layer laminate spine.

3 Robot Specifications

All key physics-based parameters for the proposed quadruped robot are collected in Table 2. Values are expressed in SI units and are drawn from biological scaling, material data, or simple calculations as indicated.

Table 2: Specifications table for the proposed quadruped robot.

Parameter	Symbol	Unit	Value / Range	Source
Total mass	m	kg	0.25	Measured
Body length (shoulder-hip)	L_{body}	m	0.25	Geometry, scaled from cat data
Target walking speed	v	m/s	0.2–0.3	Design goal
Stride length	L_{stride}	m	0.20–0.25	Design goal
Step frequency	f_{step}	Hz	1.0–1.5	Design goal
Peak vertical GRF per leg	$F_{z,\text{max}}$	N	3–5	Scaled from cat GRF data
Body weight	$W = mg$	N	≈ 2.45	Calculated from mass
Peak COM vertical acceleration	a_z	m/s ²	≈ 6.2	Calculated (this work)
Max spine pitch (front-rear)	θ_{max}	rad	≈ 0.17	Design limit
Target spine torsional stiffness	k_{θ}	N m/rad	≈ 1.4	Calculated (this work)
Laminate link modulus (FR4)	E_{link}	GPa	15–25	Datasheet / SCM literature
Flexure modulus (Kapton)	E_{flex}	GPa	2.5–3.0	Datasheet / SCM literature
Laminate stack	–	–	5-layer (rigid–adhesive–flexure–adhesive–rigid)	SCM process [1, 5]

The values in Table 2 combine biological scaling and conservative design choices. The peak ground reaction forces are scaled from biological values but slightly overestimated to provide a safety margin. The target walking speed, stride length, and step frequency are reduced relative to biology to reflect the limitations of small actuators and friction in laminate joints. The spine pitch limit and corresponding torsional stiffness are design targets chosen to allow visible bending while maintaining a strong passive restoring torque from the flexure-based joints.

4 Physical Mechanism Prototypes

4.1 Paper Mechanism Prototype

To develop intuition for the articulated spine, we first built a paper mechanism using slit-and-fold joints. Figure 1 shows the resulting model in several configurations. The prototype consists of two large prismatic frames representing the front and rear body segments and a central spine module built from folded triangular facets. All joints are implemented as thin paper folds, which act as flexible hinges.

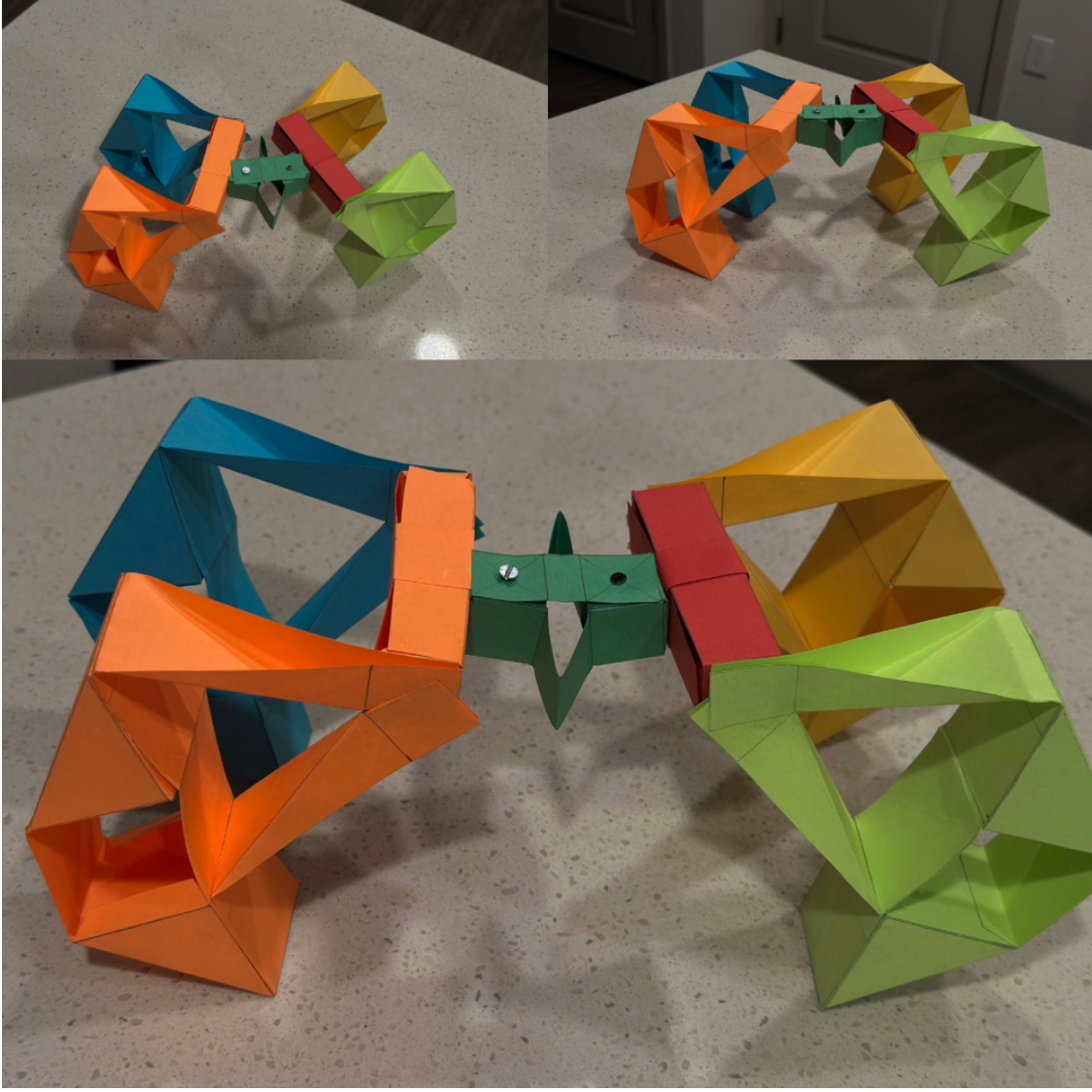


Figure 1: Paper prototype of the articulated spine mechanism in multiple configurations. The colored prismatic frames represent the front and rear body segments. The central colored elements form a spatial Sarrus-like linkage whose folded joints behave as passive hinges. By manually rotating the outer frames relative to each other, the spine bends while remaining vertically supportive, illustrating the intended behavior of the laminate spine in the final robot.

This paper analogue validates the qualitative motion of the Sarrus-based spine and helps identify locations where flexures and rigid links should be reinforced in the laminate design.

5 Mechanism Geometry and Kinematic Diagram

The physical spine module is a six-link Sarrus-based mechanism between two rigid platforms (front and rear chassis). For clarity, we draw a planar kinematic diagram in the sagittal plane, shown in Fig. 2. The ground link of length l_0 connects points P_6 and P_0 , which lie on the rear chassis. Two mirrored three-link chains connect P_6 and P_0 to a common point P_3 , which represents the

front chassis attachment. A central link of length l_6 connects P_6 to P_3 and enforces the parallel relationship between front and rear platforms.

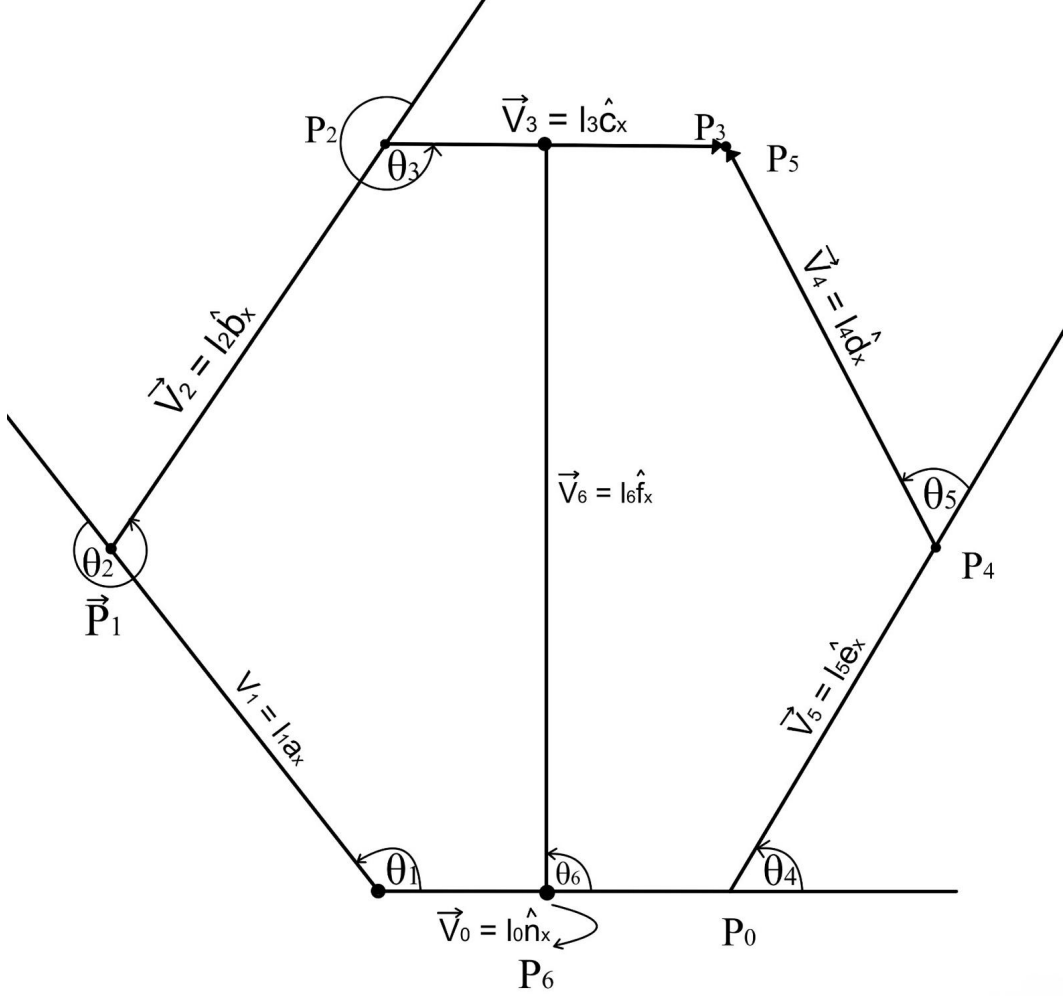


Figure 2: Planar kinematic diagram of the articulated spine. The inertial frame $\mathcal{N}$ is defined by $\{\hat{n}_x, \hat{n}_y, \hat{n}_z\}$. Rigid links are represented by vectors $\vec{V}_i = l_i(\hat{\cdot})_x$, where l_i is the length of link i and the local $(\hat{\cdot})_x$ axis is aligned with that link. Points P_0 – P_6 denote joint locations, and joint angles $\theta_1 \dots \theta_6$ are measured relative to the $\hat{n}_x$ axis. The full spatial Sarrus linkage has six moving links; for planar kinematic analysis we later approximate one branch as a 3R serial chain.

6 Estimated Goal Performance Metrics

The literature provides approximate values for body mass and ground reaction forces, but it does not directly specify several design metrics important for the articulated spine. Here we estimate (1) the peak vertical acceleration of the body during stance and (2) a target torsional stiffness for the spine.

6.1 Peak Vertical Acceleration

With total mass $m = 0.25$ kg, the robot weight is

$$W = mg = 0.25 \times 9.81 \approx 2.45 \text{ N}. \quad (1)$$

During walking, each leg experiences a peak vertical ground reaction force in the range of 3–5 N. We choose a representative value $F_{z,\max} = 4$ N for a single leg. Assuming that at the instant of peak force the net vertical force on the body is dominated by the difference between the ground reaction force and weight, Newton’s second law gives

$$F_{z,\max} - W = ma_z. \quad (2)$$

Solving for a_z ,

$$a_z = \frac{F_{z,\max} - W}{m} \approx \frac{4.0 - 2.45}{0.25} \approx 6.2 \text{ m/s}^2, \quad (3)$$

corresponding to a peak upward acceleration of roughly $0.6g$. This sets a reasonable target for the vertical dynamics that the articulated spine must accommodate without excessive deflection.

6.2 Target Torsional Stiffness of the Spine

We would like the spine to permit a maximum relative pitch angle of $\theta_{\max} \approx 10^\circ$ between the front and rear chassis under load while generating a restoring torque that tends to re-center the body. If we assume that a disturbance produces an effective moment of $\tau_{\max} = 0.25$ N m about the spine, and model the spine as a linear torsional spring,

$$\tau = k_\theta \theta, \quad (4)$$

then a target torsional stiffness is

$$k_\theta = \frac{\tau_{\max}}{\theta_{\max}} \approx \frac{0.25}{0.174} \approx 1.4 \text{ N m/rad}. \quad (5)$$

This value serves as a goal for the combined effect of all flexure joints in the articulated spine.

7 Python-Based Kinematic Model

To analyze the articulated spine, we implemented a planar kinematic model in Python using SymPy and Matplotlib. For tractability, the spatial six-link Sarrus spine is approximated as an equivalent planar 3R serial chain that connects the rear chassis attachment to the effective front chassis attachment. Each branch of the Sarrus linkage follows the same kinematics, so the serial representation captures the dominant bending motion and the mapping between endpoint forces and joint torques.

7.1 Variables, Parameters, and Frames

The model is defined in an inertial frame $\mathcal{N}$ with orthonormal basis $\{\hat{n}_x, \hat{n}_y, \hat{n}_z\}$, where $\hat{n}_x$ is horizontal and $\hat{n}_y$ is vertical. The base of the spine at the rear chassis is fixed at

$$P_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (6)$$

The equivalent spine has three rigid links of lengths l_1, l_2, l_3 with revolute joints at P_0, P_1 , and P_2 . The generalized coordinates are

$$\mathbf{q} = [\theta_1 \quad \theta_2 \quad \theta_3]^\top, \quad (7)$$

where each θ_i is a relative joint angle. The absolute orientations are

$$\alpha_1 = \theta_1, \quad \alpha_2 = \theta_1 + \theta_2, \quad \alpha_3 = \theta_1 + \theta_2 + \theta_3. \quad (8)$$

Link lengths are chosen based on the prototype: $l_1 = l_2 = 0.06$ m and $l_3 = 0.05$ m.

7.2 Forward Kinematics and Configuration Plots

Using a vector-based description, the joint positions are

$$P_1 = P_0 + l_1 \begin{bmatrix} \cos \alpha_1 \\ \sin \alpha_1 \end{bmatrix}, \quad (9)$$

$$P_2 = P_1 + l_2 \begin{bmatrix} \cos \alpha_2 \\ \sin \alpha_2 \end{bmatrix}, \quad (10)$$

$$P_3 = P_2 + l_3 \begin{bmatrix} \cos \alpha_3 \\ \sin \alpha_3 \end{bmatrix}, \quad (11)$$

where P_3 is the effective front chassis location. The Python script evaluates these expressions symbolically, then numerically for several key configurations:

- Neutral posture: $\mathbf{q}_{\text{neutral}} = [0, 0, 0]^\top$.
- Left-bending posture: $\mathbf{q}_{\text{left}} = [10^\circ, -5^\circ, 0]^\top$.
- Right-bending posture: $\mathbf{q}_{\text{right}} = [-10^\circ, 5^\circ, 0]^\top$.

Plots of these configurations show that the model produces a total spine pitch of approximately $\pm 10^\circ$, consistent with the design target.

7.3 Jacobian and Force Mapping

The endpoint position is

$$\mathbf{r}(\mathbf{q}) = \begin{bmatrix} x(\mathbf{q}) \\ y(\mathbf{q}) \end{bmatrix} = P_3(\mathbf{q}). \quad (12)$$

The planar Jacobian is

$$J(\mathbf{q}) = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \end{bmatrix}. \quad (13)$$

For static analysis, the relationship between endpoint forces and joint torques is

$$\boldsymbol{\tau} = J^\top \mathbf{F}_{\text{ee}}, \quad (14)$$

where $\mathbf{F}_{\text{ee}}$ is the force at the endpoint. Using the specifications in Sec. 3, we apply a downward load $\mathbf{F}_{\text{ee}} = [0, -4]^\top$ N and compute the resulting joint torques at each configuration. The magnitudes of these torques are within the range that can be provided by small hobby servomotors, confirming that the laminated spine is structurally reasonable.

7.4 Jacobian and Velocity Mapping

The Jacobian also relates endpoint velocities to joint velocities:

$$\dot{\mathbf{r}} = J(\mathbf{q})\dot{\mathbf{q}}. \quad (15)$$

Given a desired endpoint velocity, e.g. $\dot{\mathbf{r}} = [0, 0.05]^T$ m/s at the neutral configuration, the script uses the Jacobian pseudoinverse to compute a minimum-norm joint velocity vector $\dot{\mathbf{q}}$. This provides a first-order estimate of the required joint speeds to achieve a given vertical motion of the front chassis.

7.5 Python Implementation

The complete Python script is included in Appendix ?? and is based on the vector-kinematics examples in the course notebook. It defines all variables and constants, declares frames implicitly through link orientations, computes joint positions and Jacobians symbolically using SymPy, and visualizes the mechanism in key configurations using Matplotlib.

8 Discussion and Future Work

The articulated spine developed in this project demonstrates that a Sarrus-based laminate mechanism can provide directional stiffness and passive compliance for a cat-scale quadruped robot. The paper prototype helped refine the physical design and link layout, while the laminate prototype shows that the five-layer stack can achieve the necessary stiffness-to-compliance ratio using standard SCM materials.

The simplified planar 3R kinematic model provides a useful approximation for understanding how body-level forces map into joint torques and how desired spine motions relate to joint velocities. However, the full spatial Sarrus mechanism contains additional geometric and stiffness details that are not captured in the planar model. Future work will include:

- Extending the model to full 3D, including lateral bending and torsion.
- Integrating Jansen mechanisms on each chassis and studying the coupled leg–spine dynamics.
- Calibrating the effective spine stiffness by measuring torque versus angle for the laminate flexures and comparing to the target stiffness estimated in Sec. 6.
- Implementing closed-loop control strategies that exploit the passive compliance for perturbation rejection.

9 Conclusion

We have designed, fabricated, and modeled a laminate articulated spine for a small quadruped robot inspired by both the directional stiffness of mammalian spines and the linkage-based backbones of myriapod-style robots. The mechanism uses a Sarrus-inspired parallel linkage implemented in a five-layer laminate stack, yielding stiff links and passive spring-like joints. A paper model and a laminate prototype confirm that the mechanism can bend as desired while maintaining vertical support. A Python-based kinematic model, derived from an equivalent planar 3R chain, provides estimates of joint torques under expected ground reaction forces and joint velocities for prescribed body motions. These results establish the articulated spine as a feasible and tunable subsystem for future Jansen-leg quadruped prototypes.

References

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